

Week 2 Homework (March 12)

Due date: Mar 17 in class

1. Section 9.1: 3,4,5,6.
2. Prove that any nonempty open set $E \subset \mathbb{R}$ is the union of countably many disjoint open intervals.
3. Prove that nonempty $E \subset \mathbb{R}$ is connected iff it is an interval.